8.4.5 Decimal to Hexadecimal Conversion

In decimal to hexadecimal conversion, divide the decimal number by 16 and write down the remainder after each division. The remainders are taken in reverse order to form the hexadecimal number.

Example: Conversion of a decimal number 4538 to its equivalent hexadecimal number.

16|4538
16|283 – 10
16|17 – 11
1 – 1

Hence, $4538_{10} = 11BA_{16}$

8.4.6 Hexadecimal to Decimal Conversion

To convert the hexadecimal to its equivalent decimal number, multiply each hexadecimal digit by its weight and then add the resulting products.

Example: Conversion of a hexadecimal number of B35 to its equivalent decimal number.

The weight of B is 16^2, 3 is 16^1 and 5 is 16^0

Hence its equivalent Decimal number is

$= (Bx16^2)+(3x16^1)+(5x16^0)$

$= (11x256)+(3x16)+(5x1)$

$= 2816+48+5$

$=2869$

Hence, $B35_{16} = 2869_{10}$

8.4.7 Octal to Binary Conversion

In this, each octal digit is converted into its equivalent three digit binary form. The octal number and its equivalent three digit binary numbers are shown in the Table 8.1.

Table 8.1: Conversion of Octal into Equivalent Binary Number

Octal number	Equivalent Binary number
0	000
1	001
2	010
3	011
4	100
5	101
6	110
7	111

Example: Conversion of an octal number 43 to its equivalent binary number.

4 3

100 011

$43_8 = 100011_2$

8.4.8 Binary to Octal Conversion

The binary numbers are grouped as 3-bit from left to right. If there is any binary digit left with one or two bits then sufficient numbers of zero are added to the left most side of the binary number. Then, grouped 3-bit number is converted into an equivalent octal number.

Example: Conversion of a binary number of 010111011 to its equivalent octal number.

010 111 011

2 7 3

Hence, $010111011_2 = 273_8$

8.4.9 Hexadecimal to Binary Conversion

In this, each hexadecimal digit is converted into its equivalent four digit binary form.

The hexadecimal number and its equivalent 4 digit binary numbers are shown in the Table 8.2.

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